Date: _____

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Assignment — Assembly Language

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SSD :-

SSD stand for Solid-State Drive. it's a type of Computer storage device that uses integrated circuit assembles to store data presistintly. Unlike traditional hard disk drive (HDDs) which use spining disks and movable read/write heads. SSDs have no moving parts. This fundamental difference in design leads to several keys advantages in peformance, reliability, and energy.

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→ How an SSD work?

To understand what an SSD is, its importance to know how it function. SSD, store data using flash memory, typically NAND flash. Here a simplified explanation of how an SSD works:

→ The SSD controll manage read/write operations and data distribution.

→ When your Computer need to access data, the controller quickly retrives it from the appropriates cells.

→ For writing data, the controller finds available cells and stores the information electronically.

These process hapens extremely quickly allowing for much faster data access compared to traditional HDDs. the lacks of moving parts not only increase speed but also improves reliability and reduces power Consumption.

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Types of SSDs:-

When Considering what an SSD is its important to know that there are several types available. each with its own connection interface.

(1) SATA SSDs:- These use to same connection at traditional HDDs and are common in older systems.

(2) PCIe SSDs:- These conection directly to the motherboard's PCIe slots for faster data transfer.

(3) NVMe SSDs:- These use the PCIe interface but with an optimized protocol for even faster speeds. NVMe SSDs are becoming increasingly common in high end laptop and desktop.

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(4) mSATA SSDs:

These are smaller than standard 2.5 inch SSDs and were commonly used in ultra-thin laptop before M.2 become more prevent.

→ What is an SSD Used for?

SSDs are Used for the same Purpuse as traditional hard disk drives. but they excel in device where speed, reliblity and energy efficency are cruicial. Common Use for SSDs includs.

(1) Operating System Drives:

SSDs are ideal for installing your operating System resualting is faster boot times and improved System.

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(2) Application Storage:-

SSDs can dramatically reduce load time and improve overall performance. This is particularly noticeable in video editing software or development environments.

(3) Gaming:-

SSDs are a game-changer for games. They significantly reduce load times for game, allowing player to get into the action faster. They also improve in-game performance when loading new or assets.

(4) Laptop and Protable Devices.

The low power consumption, durability and compact size of SSD make them ideal use in laptop, tablets and protable device where energy efficiency and physical reliance are important.

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Conclusion:

Understanding what an SSD is and its benifits is crucial in today's Computing Landscape. Solid-state drives represents a Significant leap forward in storage technology, offering unparalleled speed, reliability and efficiency. As SSD price continue to fall and capacities increase. They've becoming the standard storage solution in modern Computer.

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Hard Disk:

A harddisk is an important Storage device used in Computer. it is used to store on a harddisk. can be accessed whenever it is needed.

A harddisk is called hard drive (HDD). is a secondry storage device. it store data when power is switched off.

Parts of a Harddisk.

→ Platter: where data store

→ Spindle: it store the platter at high Speed.

→ Actuator: it controal the moment of the actuator arm

→ Power Connector: it Supplies electrical power to the hard disk.

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=> Work of Hard Disk:-

Suppose you have a picture sends the data to hard disk. The platter rotates at high speed. The read write head moves over the platter to read or write the required information. The operating system manages the location of files so that they can be accessed later.

Types of Hard disk:-

HDD:-

- it use spanning magnatic platter to store data. it usually provide large storage at a lower cost.

SSD:-

- it use flash memory and has no moving parts. it is generally faster and quieter then HDD.

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Advantages of Hard Disk:

→ Provide Large storage capacity
→ Can be used for backups and general data storage.
→ Can store data permanently
→ Useful for storing large file.

Conclusion:

A hard disk is an important computer storage device. it provide permenant storage for different type of data. Although SSD are becoming more popular because of their speed, HDD are still widely used because they offer large storage capacity at relative cost.

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